" DYNAMIC MODEL OF A ONE-LINK ROBOT MANIPULATOR WITH BOTH STRUCTURAL AND JOINT FLEXIBILITY "

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Abstract

We have developed the equations which describe the motion of a robot manipulator which incorporates both structural and joint flexibility . We have focused here on the dynamic model of a one-link flexible manipulator in order to show how critical it is to incorporate joint flexibility and also to demonstrate the significance of cross coupling among state variables for small deflection. A set of decoupled dynamic equations was obtained based on the Assumed-Mode Method and on orthogonality relations. By comparison to a detailed model which does incorporate cross coupling, it was shown that it is sufficiently accurate to use the decoupled dynamic equations for small joint and structural deflections. Simulation of varying joint stiffness characteristics indicates that there is no rigid mode for the flexible joint system. As such, both structural and joint flexibility must be considered in the analysis and control of such systems.

Introduction

Robot manipulators have for some time been widely used both in hostile environments and for industrial applications. However, experience has shown that for many tasks, presently available robots are neither sufficiently accurate to be effective nor fast enough to be economical. The requirements for higher speeds and better system performance make it a necessity to consider the dynamic effects of both structural flexibility and joint flexibility in the design of manipulators [1-2].

Much attention has been paid to the structural flexibility of robots. The dynamic behavior of flexible manipulators has been investigated by a number of researchers. Their studies can be categorized into two groups: (1) those that consider flexible structures with rigid joints; and (2) those that consider rigid structures with flexible joints.

The interest in this area has been further stimulated by the work on large space structures (LSS) for NASA. This program has been a source of new challenges in modeling and control of large flexible structures with very low natural damping operating under zero gravity force [3]. Several different approaches have been considered in modeling and control of the elastic link manipulators. Early work on the modeling and control of a single flexible link device was reported by Mirro [4]. Book, Whitney [5-6] extended this work to a two-link planar elastic arm using a transfer matrix method. The analysis of the elastic-dynamic behavior of a planar mechanical linkage system was considered in great detail using a variety of dynamic equation formulation techniques summarized by Lowen and Jandrasits [7]. Thompson [8] treated the mechanism as continuous systems with the links considered only as elastic elements. Dubowsky, Gardner [9] and Midha etc [10] treated the mechanism as a discrete system, where the elastic links are replaced by a finite number of masses and springs. A topological chain model for robot manipulator representation was introduced by Dubowsky and Gardner [9]. Sunada and Dubowsky [1] developed a Component Mode Synthesis procedure with a finite element method applied to a flexible, spatial kinematic system. None of the control schemes based on above dynamic models have used end-point feedback control. The lack of a reasonable sensor of the world position of the end-effector makes it even more different to come up with a design for a high performance controller for flexible manipulators. An end-point control method was first introduced by Cannon and Schmitz [11]. Their experiment results indicate that good closed-loop performance was achieved. However due to there sensor design, they were limited to motion in the plane. We are developing a new system for full 6 DOF tracking of end effector motion [12]. However, key issues must still be addressed in order to establish an appropriate model for a flexible manipulator.

The limitation for all the methods mentioned above is that they were based on the assumption of a rigid joint. It has been determined experimentally that certain elasticity exists in the driven system. Sweet and Good [13] reported on the torsional elasticity found in the driven system of a GE P-50 robot. Our results further show that the joint flexibility can cause significant effects on the deflection of flexible arms.

A few researchers have investigated the dynamic behavior of robotic manipulators considering rigid links and an elastic joint. Marino, Nicosia [2,14] have used a singular perturbation formulation [15] to derive a dynamic model for a robot manipulator with a weakly elastic joint. Khorasaui and Spong [16] utilized the concept of the invariant manifold to represent the dynamics of the slow subsystem. The premise behind this work is to model the flexible links as a chain of rigid sublinks interconnected by elastic joints. To get good results, one has to use a large number of sublinks with elastic joints to represent a multi-link system. This leads to a high order system model, which further complicates its control design.

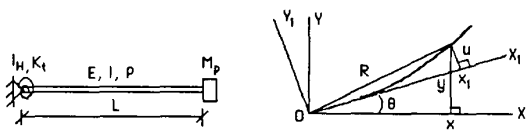
This paper will describe two approaches to dynamic modeling of a single degree robot manipulator which includes both structural flexibility and joint flexibility. The simplified model leads to a set of decoupled dynamic equations. A more detailed model leads to a set of cross-coupled dynamic equations. We will show that these two models are very close for small deflection and small angular rotation. The results from our simulations further demonstrate the importance of incorporating both structural and joint flexibility in the dynamic model. The conclusions that we will present is that a

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simplified model is sufficient but that both structural and joint flexibility must be included.

Dynamic Model based on decoupling

The system studied here includes a uniform distributed pinned-free beam of length L in the horizontal plane as shown in Fig.1. Where E, I, ρ, and M_p are Young's modulus, the moment of inertia of the cross-sectional area, the density (mass per unit length) and payload. I_H is the hub inertia. The flexible joint is modeled as shown by a torsional spring with stiffness K_t. The angular rotation, θ, is the sum of the angular displacement due to the hub rotation, θ_H, and the angular displacement due to the flexibility of the joint, θ_K. XOY is the original coordinate system, while X_1OY_1 is the coordinate system after an angular rotation θ. x and x_1 are the distance between a point on the beam and the origin, before and after the angular rotation θ.



(a) Schematic (b) Terminology

Fig.1 A single degree flexible manipulator

For small angular rotation θ, and deflection u ($u<<x_1$), we can assume

$$y = x\theta + u \tag{1}$$

Then the kinetic energy T and potential energy V for such a system with a payload M_p can be written as

$$T = \frac{1}{2} I_H \dot{\theta}^2 + \frac{1}{2}\int_0^l \left(\frac{\partial u}{\partial t} + x\dot{\theta}\right)^2 \rho dx + \frac{1}{2} M_p \left(\frac{\partial u}{\partial t} + x\dot{\theta}\right)^2_{x=0} \tag{2}$$

$$V = \frac{1}{2}\int_0^l EI \left(\frac{\partial^2 u}{\partial x^2}\right)^2 dx + \frac{1}{2} K_t \theta^2 \tag{3}$$

The nonconservative work for input torque T_{in} is

$$W = T_{in}\theta \tag{4}$$

By using Hamilton's Principle [18], we have

$$\delta\int_{t1}^{t2} (T - V + W)\, dt = \delta\int_{t1}^{t2} \left[\frac{1}{2} I_H \dot{\theta}^2 + \frac{1}{2}\int_0^L \left(\frac{\partial u}{\partial t} + x\dot{\theta}\right)^2 \rho dx + \frac{1}{2} M_p \left(\frac{\partial u}{\partial t} + x\dot{\theta}\right)^2_{x=L} - \frac{1}{2}\int_0^L EI \left(\frac{\partial^2 u}{\partial x^2}\right)^2 dx - \frac{1}{2} k_t \theta^2 + T_{in}\theta\right] dt \tag{5}$$

The rotatory inertia and shear deformation are more pronounced at the high frequencies and more influencial on the higher modes [17]. Since the later results show that we can use the first two modes in the dynamic modeling, we ignoring the rotatory inertia and shear deformation effects. After mathematical manipulation of Eq. 5, we have the beam equation

$$EI \frac{\partial^4 u}{\partial x^4} + \rho \frac{\partial^2 u}{\partial t^2} = -\rho x \ddot{\theta} \tag{6}$$

with the four boundary conditions:

$$\begin{aligned} &EI \left(\frac{\partial^2 u}{\partial x^2}\right)_{x=0} - I_H \ddot{\theta} - k_t \theta + T_{in} = 0 \\ &u_{x=0} = 0 \\ &EI \left(\frac{\partial^2 u}{\partial x^2}\right)_{x=L} = 0 \\ &EI \left(\frac{\partial^3 u}{\partial x^3}\right)_{x=L} = M_p \left(\frac{\partial^2 u}{\partial t^2} + x\ddot{\theta}\right)_{x=L} \end{aligned} \tag{7}$$

For $y(x,t) = u(x,t) + x\,\theta(t)$, Eqs. (6) and (7) become

$$EI \frac{\partial^4 y}{\partial x^4} + \rho \frac{\partial^2 y}{\partial t^2} = 0 \tag{8}$$

and

$$\begin{aligned} &EI \left(\frac{\partial^2 y}{\partial x^2}\right)_{x=0} - I_H \ddot{\theta} - k_t \theta + T_{in} = 0 \\ &y_{x=0} = 0 \\ &EI \left(\frac{\partial^2 y}{\partial x^2}\right)_{x=L} = 0 \\ &EI \left(\frac{\partial^3 y}{\partial x^3}\right)_{x=L} = M_p \left(\frac{\partial^2 y}{\partial t^2} + x\ddot{\theta}\right)_{x=L} \end{aligned} \tag{9}$$

Using the Assumed-Mode method [18], the solution of the flexible motion can be represented as the linear combination of admissible functions $\Phi_i(x)$ multiplied by time-dependent generalized coordinates $q_i(t)$, where the admissible function $\Phi_i(x)$ should satisfy all the essential boundary conditions, that is

$$y = \sum_{i=1}^{n} \Phi_i q_i \tag{10}$$

In order to study the orthogonality of natural modes, the external torque T_{in} is set to zero. The orthogonality relationship of the second derivatives, which is true for a system with only structural flexibility, will not hold here. Instead, we have

$$\int_0^L \rho \Phi_i \Phi_j dx + I_H \dot{\Phi}_i(0)\dot{\Phi}_j(0) + M_p \Phi_i(L)\Phi_j(L) = \begin{cases} I_a \text{ (constant)} & \text{for } i = j \\ 0 & \text{for } i \neq j \end{cases} \tag{11}$$

and

$$\int_0^L EI\, \ddot{\Phi}_i \ddot{\Phi}_j\, dx = \begin{cases} \omega_i^2 I_a - k_t [\dot{\Phi}_i(0)]^2 & \text{for } i = j \\ -k_t \dot{\Phi}_i(0)\, \dot{\Phi}_j(0) & \text{for } i \neq j \end{cases} \tag{12}$$

Substituting Eq. (10) into Eqs. (2), (3) and using the orthogonality relations Eqs. (11) and (12); the expressions for kinetic and potential energy become

$$\begin{aligned} T &= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} \dot{q}_i \dot{q}_j \left[\int_0^L \rho \Phi_i \Phi_j dx + I_H \dot{\Phi}_i(0)\dot{\Phi}_j(0) + M_p \Phi_i(L)\Phi_j(L)\right] \\ &= \frac{1}{2}\delta_{ij} I_a \sum_{i=1}^{n} \dot{q}_i^2 \end{aligned} \tag{13}$$

$$\begin{aligned} V &= \frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n} q_i q_j \int_0^L EI\, \ddot{\Phi}_i \ddot{\Phi}_j\, dx + \frac{1}{2} k_t \sum_{i=1}^{n}\sum_{j=1}^{n} \dot{\Phi}_i(0)\, \dot{\Phi}_j(0)\, q_i q_j \\ &= \left[\frac{1}{2}\delta_{ij} I_a \sum_{i=1}^{n} \omega_i^2 q_i^2 - \frac{1}{2} k_t \sum_{i=1}^{n}\sum_{j=1}^{n} \dot{\Phi}_i(0)\dot{\Phi}_j(0) q_i q_j\right] + \frac{1}{2} k_t \sum_{i=1}^{n}\sum_{j=1}^{n} \dot{\Phi}_i(0)\dot{\Phi}_j(0) q_i q_j \\ &= \frac{1}{2}\delta_{ij} I_a \sum_{i=1}^{n} \omega_i^2 q_i^2 \end{aligned} \tag{14}$$

and

$$W = T_{in}\theta = T_{in} \sum_{i=1}^{n} \Phi_i(0)\, q_i \tag{15}$$

Since the Lagrangian formulation of robot manipulator dynamics is desirable from a control point of view [19], the final dynamic equations based on the Lagrange's method can be obtained in the state space form :

$$\dot{X} = A\,X + B\,U_{in} \tag{16}$$

where U_{in} is a vector of input T_{in}, and

$$X = \begin{bmatrix} q_0 \\ \dot{q}_0 \\ q_1 \\ \dot{q}_1 \\ \vdots \\ q_n \\ \dot{q}_n \end{bmatrix};\quad A = \begin{bmatrix} 0 & 1 & & & & \\ 0 & 0 & & & & \\ & & 0 & 1 & & \\ & & -\omega_1^2 & -2\zeta_1\omega_1 & & \\ & & & & \ddots & \\ & & & & 0 & 1 \\ & & & & -\omega_n^2 & -2\zeta_n\omega_n \end{bmatrix};\quad B = \frac{1}{I_s}\begin{bmatrix} 0 \\ 1 \\ 0 \\ \Phi_1(0) \\ \vdots \\ 0 \\ \Phi_n(0) \end{bmatrix} \tag{17}$$

The selected system outputs are the tip position, the angular displacement and the strain, to be measured respectively by an end point sensor, an angle sensor, and strain gauges mounted immediately adjacent to the joint on the side of beam. The output vector, Z, can be expressed by state variables as follows:

$$Z = \begin{bmatrix} y(L) \\ \theta \\ \epsilon \end{bmatrix} = \begin{bmatrix} \Sigma\, \Phi_i(L) q_i \\ \Sigma\, \Phi_i(0) q_i \\ \Sigma\, \frac{\delta}{2}\, \Phi_i(0) q_i \end{bmatrix} = [C]\,X \tag{18}$$

where

$$C = \begin{bmatrix} L & 0 & \Phi_1(L) & 0 & \cdots & \Phi_n(L) & 0 \\ 1 & 0 & \Phi_1(0) & 0 & \cdots & \Phi_n(0) & 0 \\ 0 & 0 & \frac{\delta}{2}\Phi_1(0) & 0 & \cdots & \frac{\delta}{2}\Phi_n(0) & 0 \end{bmatrix} \tag{19}$$

Dynamic Model Incorporating Cross-Coupling:

A more detailed model of this single degree system can be shown in Fig. 2, where the angular displacement due to the hub rotation, θ_H, and the angular displacement due to the flexibility of the joint, θ_K, are separated out. u_d here is the deflection of the beam in this detailed model.

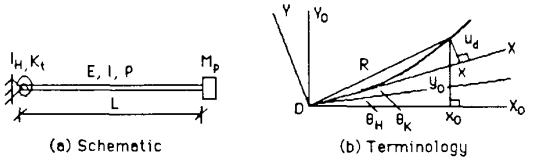


Fig. 2 A single degree flexible system

Using the transformation matrix, we have

$$\begin{bmatrix} x_0 \\ y_0 \end{bmatrix} = \begin{bmatrix} C(\theta_H + \theta_k) & -S(\theta_H + \theta_k) \\ S(\theta_H + \theta_k) & C(\theta_H + \theta_k) \end{bmatrix} \begin{bmatrix} x \\ u_d \end{bmatrix} \tag{20}$$

The kinetic energy and potential energy then become

$$T = \frac{1}{2} I_H \dot{\theta}_H^2 + \frac{1}{2}\int_0^L [\,(x^2 + u_d^2)(\dot{\theta}_H + \dot{\theta}_k)^2 + \dot{u}_d^2 + 2x\dot{u}_d(\dot{\theta}_H + \dot{\theta}_k)\,]\,\rho dx$$

$$+ \frac{1}{2} M_p [\,(L^2 + u_d^2(L))(\dot{\theta}_H + \dot{\theta}_k)^2 + \dot{u}_d^2(L) + 2L\dot{u}_d(L)(\dot{\theta}_H + \dot{\theta}_k)\,] \tag{21}$$

$$V = \frac{1}{2}\int_0^L EI\left(\frac{\partial^2 u_d}{\partial x^2}\right)^2 dx + \frac{1}{2} k_t \left(\frac{\partial u_d}{\partial x}\right)^2_{x=0} + \frac{1}{2} k_t \theta_k^2 \tag{22}$$

Rewriting Eq. (10) with u_d replacing y and taking the first two flexible modes, we have

$$u_d = \Phi_1(x)\, q_1(t) + \Phi_2(x)\, q_2(t) \tag{23}$$

The final equations for the cross-coupled system based on Lagrange's method can then be written as

$$[A_d]\,\ddot{X} + [B_d]\,\dot{X} + \underline{NL} = \underline{Q} \tag{24}$$

where $\underline{Q}$ is a generized force vector, which is similar to $\underline{U_{in}}$ in Eq. (16), and

$$X = \begin{bmatrix} \theta_H \\ \theta_K \\ q_1 \\ q_2 \end{bmatrix}\quad A_d = \begin{bmatrix} A_{11} + I_H & A_{11} & A_{13} & A_{14} \\ A_{11} & A_{11} & A_{13} & A_{14} \\ A_{13} & A_{13} & \int_0^L \rho\Phi_1^2 dx + M_p\Phi_1^2(L) & 0 \\ A_{14} & A_{14} & 0 & \int_0^L \rho\Phi_2^2 dx + M_p\Phi_2^2(L) \end{bmatrix}$$

$$B_d = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & K_t & 0 & 0 \\ 0 & 0 & \int_0^L EI\ddot{\Phi}_1 dx + K_t\dot{\Phi}_1^2(0) & 0 \\ 0 & 0 & 0 & \int_0^L EI\ddot{\Phi}_2 dx + K_t\dot{\Phi}_2^2(0) \end{bmatrix}\quad Q = \begin{bmatrix} T_H \\ T_K \\ 0 \\ 0 \end{bmatrix}$$

and

$$A_{11} = \int_0^L \rho x_1^2 dx + M_p L^2;\quad A_{13} = \int_0^L \rho x_1 \Phi_1 dx + M_p L \Phi_1(L);$$

$$A_{14} = \int_0^L \rho x_1 \Phi_2 dx + M_p L \Phi_2(L);$$

$$NL1 = \Big[\int_0^L (\Phi_1^2 q_1^2 + \Phi_2^2 q_2^2 + 2\Phi_1\Phi_2 q_1 q_2)\rho dx + M_p(\Phi_1^2(L)q_1^2 + \Phi_2^2(L)q_2^2$$
$$+ 2\Phi_1(L)\Phi_2(L)q_1 q_2)\Big](\ddot{\theta}_H + \ddot{\theta}_k) + \int_0^L 2\rho(\Phi_1^2 q_1\dot{q}_1 + \Phi_2^2 q_2\dot{q}_2 + \Phi_1\Phi_2 q_1\dot{q}_2$$
$$+ \Phi_1\Phi_2\dot{q}_1 q_2)(\dot{\theta}_H + \dot{\theta}_k)dx + 2M_p\big[\Phi_1^2(L)q_1\dot{q}_1 + \Phi_2^2(L)q_2\dot{q}_2 + \Phi_1(L)\Phi_2(L)q_1\dot{q}_2$$
$$+ \Phi_1(L)\Phi_2(L)\dot{q}_1 q_2\big](\dot{\theta}_H + \dot{\theta}_k)$$

$$NL2 = NL1$$

$$NL3 = -\int_0^L (\Phi_1^2 q_1 + \Phi_1\Phi_2 q_2)(\dot{\theta}_H + \dot{\theta}_K)^2 \rho dx + M_p(\Phi_1^2(L)q_1 + \Phi_1(L)\Phi_2(L)q_2)(\dot{\theta}_H + \dot{\theta}_K)^2$$

$$NL4 = -\int_0^L (\Phi_2^2 q_2 + \Phi_1\Phi_2 q_1)(\dot{\theta}_H + \dot{\theta}_K)^2 \rho dx + M_p(\Phi_2^2(L)q_2 + \Phi_1(L)\Phi_2(L)q_1)(\dot{\theta}_H + \dot{\theta}_K)^2 \tag{25}$$

Subtracting the second row of Eq. (24) from the first row, we get

$$I_H\ddot{\theta}_H - k_t\theta_K = T_H - T_K \tag{26}$$

By linearizing Eq. (24), all the nonlinear terms, $\underline{NL}$ (NL1 through NL4), disappear. Since q_1 and q_2 are a combination of sine and cosine functions with natural frequencies ω_1 and ω_2, we can conclude that θ_H and θ_K should contain the same sine and cosine functions with frequency ω, which is a function of ω_1 and ω_2. Therefore, we have

$$\theta_K = k_1\theta_H + k_0 \tag{27}$$

where K_1, K_0 are related the beam equations and

$$\theta = \theta_H + \theta_K = f_1(\theta_H) = f_2(\theta_K) \tag{28}$$

Thus it stands to reason that θ used in the simple decoupled model can be used instead of θ_H and θ_K from the cross-

coupled model. If one further restricts oneself to small structural and joint deflections, then the decoupled equations are equivalent to the cross-coupled equations.

Results

In order to illustrate the significance of the dynamic models developed in this paper, simulation of a single degree of freedom manipulator, including both a flexible structure and a flexible joint, are shown below. A comparison among different joint flexibilities is made in order to demonstrate the importance of incorporating joint flexibility into the dynamic model. The examples in this paper are based on the decoupled dynamic model incorporating only the first two flexible modes. To simplify the computation, the following parameters are used :

$$\beta^4 = \frac{\rho}{EI}\omega^2,\ \lambda = \beta L,\ I_B = \frac{1}{3}\rho L^3,\ \varepsilon = \frac{I_H}{3I_B},\ \eta = \frac{M_p L^2}{3I_B},\ K_c = \frac{K_t}{(EI/L)}. \quad (29)$$

We assume that the beam is made of aluminum. These parameters would be representative of a 5 cm x 0.2 cm uniform cross section beam : E = 7E10 N/m^2, ρ = 0.277 kg/m, I = 0.33E-10 m^4, L = 1 m, I_T = 0.5 kg-m^2 (for payload M_p = 0) or I_T = 0.8 kg-m^2 (for payload M_p = 0.277 kg). We also introduce the viscous damping coefficient ζ = 0.015 to each mode of the system. From Eqs. (9), (10) and (29), we have the characteristic equation

$$D(\lambda) = (-\varepsilon\lambda^4 + K_c)\,[\,1 + \cos\lambda\,\mathrm{ch}\lambda + \eta\lambda\,(\cos\lambda\,\mathrm{sh}\lambda - \sin\lambda\,\mathrm{ch}\lambda)] - \lambda\,(\sin\lambda\,\mathrm{ch}\lambda - \cos\lambda\,\mathrm{sh}\lambda + 2\eta\lambda\sin\lambda\,\mathrm{sh}\lambda) = 0 \quad (30)$$

Table 1 gives the system mode parameters λ_i based on Eq. (30). We find from Table 1 that there is no zero mode (i.e. rigid mode) for a flexible joint system ($K_c \neq 0$).

Table 1. System Mode Parameters λ_i

ε	K_c	η	λ_i				
			mode 0	1	2	3	4
0.05	0	0	0	2.967543	4.897182	7.897060	11.010750
		1	0	2.630312	4.338585	7.190448	10.275320
	1	0		1.218520	3.090267	4.904642	7.897284
		1		0.865716	2.789944	4.356291	7.190885
	5	0		1.603735	3.486017	4.939261	7.898203
		1		1.100850	3.217521	4.440913	7.192700

A package for computer aided design of control system, MAControl [20], was used to simulate the open loop system response for a unit step input in torque. The simulation results are shown in Figures 3 through 14 . Figures 3 to 6 give the first four modes of the flexible arms for three different joint stiffness coefficients. The first two mode shapes for these flexible arms are plotted in Fig. 7 and 8. Fig. 9 shows the deflection of an entire flexible arm (K_c = 1) at t = 0.1, 0.3, 0.5 second, respectively. A comparison of the deflections for three different flexible arms at t = 0.3 sec is given in Fig. 10. In both cases, the deflection is computed based on the summation over two modes $u = \Phi_1 q_1 + \Phi_2 q_2$. Figures 11 to 13 plot the output variables; tip position, angular rotation, and strain vs. time. The tip deflection, the difference between the desired and actual end-effector position due to flexilibities, is plotted in Fig.14. The effect of the joint flexibility is shown in Fig.10 and 11. The deviation of curve 2 (K_c = 1) from curve 1 (K_c = 0, rigid joint) in Fig.11 is approximately 50%. For curve 3 (K_c = 5), the deviation from curve 1 increases to 80% at t = 0.6 sec. Since a large amount of the input energy is stored in the spring-like flexible joint, the deflection of the beam with the flexible joint is much smaller than the one with the rigid joint, although it does have significantly more oscillatory behavior.

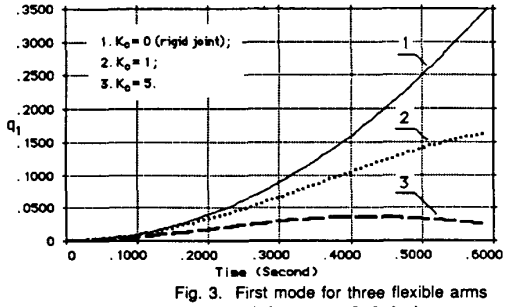


Fig. 3. First mode for three flexible arms (case 1: λ_0; case 2, 3: λ_1.)

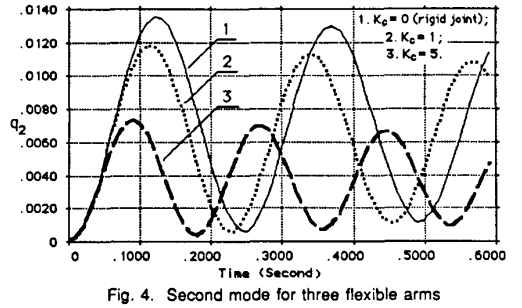


Fig. 4. Second mode for three flexible arms (case 1: λ_1; case 2, 3: λ_2.)

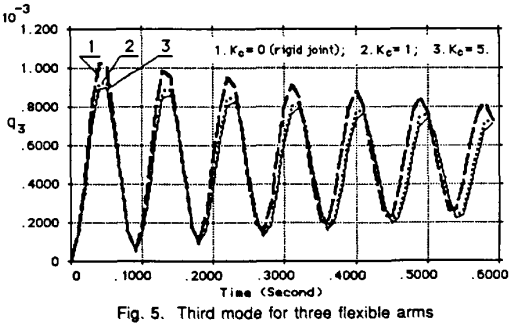


Fig. 5. Third mode for three flexible arms (case 1: λ_2; case 2, 3: λ_3.)

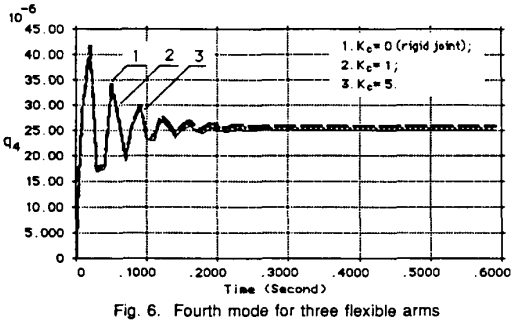


Fig. 6. Fourth mode for three flexible arms (case 1: λ_3; case 2, 3: λ_4.)

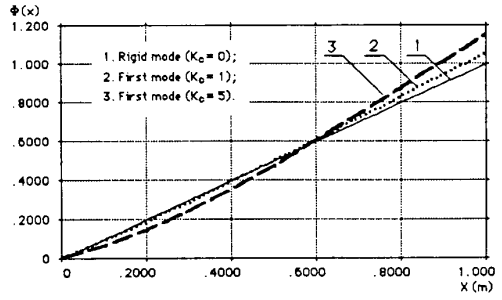


Fig. 7. First mode shape for three flexible arms (case 1: λ_0; case 2, 3: λ_1.)

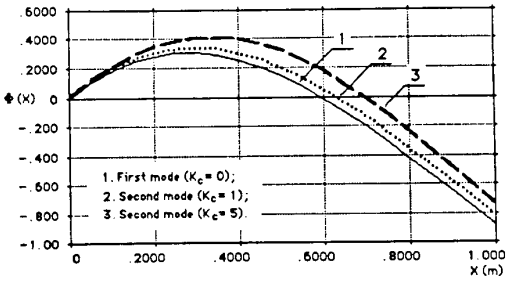


Fig. 8. Second mode shape for three flexible arms (case 1: λ_1; case 2, 3: λ_2.)

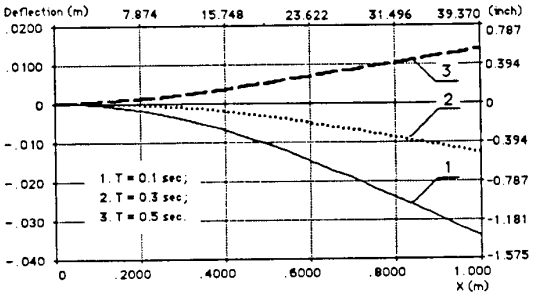


Fig. 9. Deflection of the flexible arm (K_c = 1)

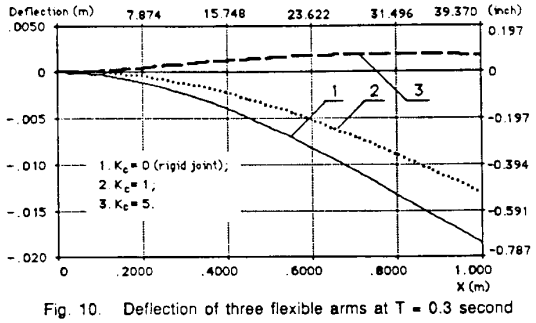


Fig. 10. Deflection of three flexible arms at T = 0.3 second

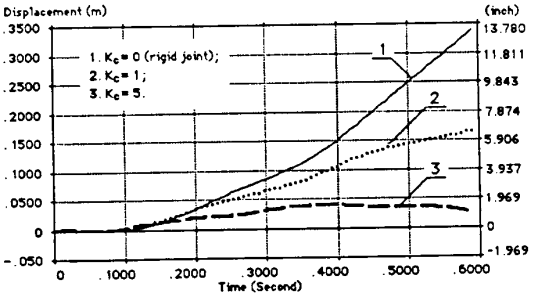


Fig. 11. Tip position for three flexible arms (measured from initial coordinate system)

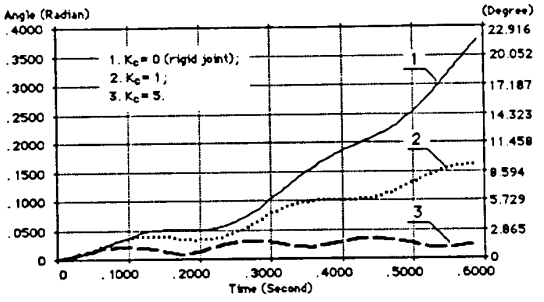


Fig. 12. Angle θ for three flexible arms

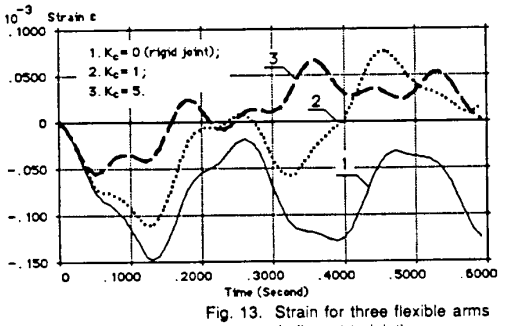


Fig. 13. Strain for three flexible arms (adjacent to joint)

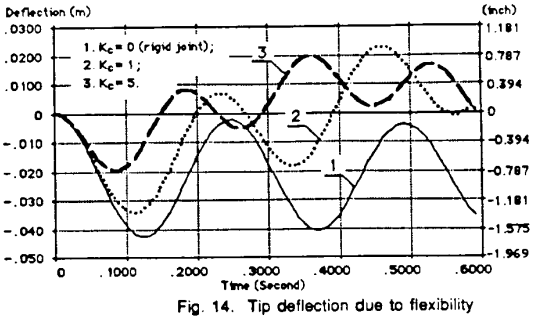


Fig. 14. Tip deflection due to flexibility

Conclusion

The motivation behind this work was (a) to show that a simpler decoupled set of equations sufficiently describes the system behavior, (b) to determine the dynamic equations describing a manipulator incorporating both joint and structural flexibility and (c) to show the need to include both joint and structural flexibility in analysing the flexural effects in manipulators. The analytic study of a detailed model incorporating cross-coupling demonstrates that the dynamic equations based on a decoupling model are accurate enough when angular q and deflection u are small. Simpler decoupled dynamic equations are much more convenient for designing controllers. The results in Table 1 show that there will be no rigid mode for a flexible joint system (i.e. $K_c > 0$). Our results for the first four modes of the generalized coordinates (Figures 3 to 6) show that the amplitudes of the higher modes are very small compared to the first two modes (λ_0, λ_1 for the rigid joint; and λ_1, λ_2 for the flexible joint). This indicates that a reduced model which only considers the first two modes can be used for future controller design. The results for the output variables (Figures 7-14) show that both structural and joint flexibility must be included in the dynamic model. The deflections vary greatly for the arms with different joint flexibilities. The oscillatory periods for tip vibration are >> 5 seconds for the rigid joint ($K_c = 0$), 1.45 and 0.8 seconds for flexible the joint ($K_c = 1, 5$), respectively. The inclusion of joint flexibility cuts down on the total deflections, but increases the oscillatory behavior of the beam. If the joint flexibilities are excluded, large errors would result; in particular, oscillatory behaviors would increase significantly. As a result, a controller design based on only a rigid joint would probably be unsuccessful.

We have only considerd a single link, single degree of freedom system. Since the orthogonal property of the second derivatives does not apply for a flexible joint system, the dynamic equations of a multi-link system will not be decoupled. Nevertheless, the methods presented here for the decoupled model can still be used to simplify the dynamic equations of the multi-link system. Work is continuing on an experimental verification of the single link flexible manipulator and on the further development of a dynamic model of a two-link manipulator with flexible structure and joints.

Acknowledgments

This work was supported by the National Science Foundation PYI Grant #DMC-8351827, and by a grant from the Minnesota Supercomputer Institute.

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